

TYNDALL AFB, FL, USA

WMO: 747750

Lat: 30.067N

Lon: 85.583W

Elev: 17

StdP: 14.69

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13846

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.5	35.4	16.1	12.4	36.9	20.8	15.6	40.5	24.9	61.6	22.8	58.9	7.9	0	0.412

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	91.2	78.5	90.2	78.5	89.1	78.4	82.2	87.3	81.2	86.7	80.3	86.1	10.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.9	160.7	85.0	79.5	153.1	84.6	78.6	148.6	84.1	45.7	87.2	44.6	86.8	43.7	86.3	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.8	18.5	16.6	DB	25.2	95.1	4.3	3.2	22.1	97.5	19.6	99.4	17.1	101.2	14.0	103.5
			WB	22.3	83.5	4.2	1.4	19.3	84.5	16.8	85.3	14.5	86.0	11.5	87.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	69.0	52.9	56.9	61.5	67.7	74.8	80.2	82.0	81.9	79.1	71.3	61.8	56.5
	DBStd	11.98	9.03	8.20	7.13	5.45	4.29	2.88	2.45	2.32	3.46	6.17	7.36	8.72
	HDD50	154	74	29	8	0	0	0	0	0	0	0	6	37
	HDD65	1264	385	241	148	37	2	0	0	0	0	20	146	285
	CDD50	7070	163	222	365	531	770	907	991	990	874	661	358	238
	CDD65	2708	9	15	41	118	307	457	526	525	424	216	48	22
	CDH74	24817	9	24	99	517	2396	4679	5920	5660	3782	1462	226	43
	CDH80	8445	0	0	1	38	551	1630	2345	2211	1319	332	18	0
(14) Wind	WSAvg	8.0	8.6	8.7	9.2	8.9	8.3	7.8	7.2	6.7	7.0	7.4	7.4	8.2
(15) Precipitation	PrecAvg	53.2	4.4	4.4	5.2	3.7	2.3	4.6	5.7	6.8	5.6	3.3	3.5	3.7
	PrecMax	73.3	17.3	10.6	10.6	9.9	7.9	9.3	10.0	12.8	23.2	8.7	9.6	9.2
	PrecMin	36.7	1.4	0.8	0.0	0.8	0.0	0.6	1.8	2.8	0.1	0.6	0.5	0.8
	PrecStd	10.1	3.5	2.7	2.8	2.7	2.0	2.3	2.2	2.8	4.6	2.1	2.0	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.2	77.2	79.1	83.5	88.8	92.8	93.7	93.1	92.4	88.6	82.3	77.7
		MCWB	68.7	68.3	68.7	70.9	73.9	78.4	79.3	78.7	76.2	75.5	72.5	71.0
	2%	DB	72.4	73.7	76.8	81.0	86.3	90.1	90.8	91.1	90.1	86.0	79.2	75.0
		MCWB	68.0	66.8	68.0	70.9	74.7	78.6	79.1	79.1	76.6	75.2	71.1	69.6
	5%	DB	69.7	71.7	74.8	79.1	84.7	88.4	89.8	90.1	88.3	83.8	76.8	72.5
		MCWB	66.1	65.6	66.6	70.2	74.7	78.2	79.1	79.2	76.8	73.6	69.3	68.6
	10%	DB	66.8	68.8	72.5	77.1	83.3	87.2	88.3	88.5	86.7	81.5	74.1	70.1
		MCWB	63.7	64.4	65.8	69.5	74.2	77.8	78.8	78.9	76.8	72.5	67.3	66.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.5	70.8	72.9	76.7	79.5	82.1	83.3	83.7	81.9	80.3	76.4	73.3
		MCDB	73.3	74.1	76.3	79.4	83.9	87.8	88.2	88.3	85.3	83.8	80.2	75.6
	2%	WB	69.1	69.2	70.9	74.5	78.0	80.8	81.9	82.3	80.2	78.4	73.5	71.2
		MCDB	70.9	71.9	74.1	78.2	83.1	86.9	87.6	87.4	85.4	82.4	76.9	73.7
	5%	WB	66.8	67.6	69.4	73.1	76.9	79.9	80.9	81.3	79.1	76.6	71.4	69.4
		MCDB	68.6	69.9	72.1	76.7	82.2	86.0	86.9	86.7	85.0	81.2	74.9	71.7
	10%	WB	64.3	65.8	67.7	71.5	75.8	78.9	80.1	80.3	78.1	74.7	69.2	67.4
		MCDB	66.2	68.0	70.7	75.2	81.0	85.0	86.2	86.1	84.1	79.4	73.1	69.5
(35) Mean Daily Temperature Range	5% DB	MDBR	18.2	17.4	17.1	16.7	15.7	13.3	13.0	13.2	14.8	17.8	19.5	17.5
		MCDBR	16.9	17.2	17.3	17.1	16.6	14.8	14.4	14.9	15.9	17.7	18.9	16.9
		MCWBR	12.6	11.8	10.4	9.7	8.2	6.8	6.6	6.8	6.7	9.1	11.8	12.5
	5% WB	MCDBR	14.4	14.0	13.7	13.8	14.0	13.0	12.9	13.0	13.7	14.1	16.1	15.4
		MCWBR	11.8	11.3	10.1	9.8	8.1	6.8	6.8	7.1	6.9	9.1	12.3	13.1
(40) Clear-Sky Solar Irradiance	taub		0.339	0.352	0.368	0.395	0.437	0.467	0.495	0.490	0.436	0.385	0.356	0.344
	taud		2.531	2.499	2.456	2.380	2.296	2.268	2.204	2.219	2.347	2.470	2.487	2.530
	Ebn at Noon		284	290	291	285	272	263	255	255	266	274	275	276
	Edh at Noon		28	31	35	39	42	43	46	45	38	32	29	27
(44) All-Sky Solar Radiation	RadAvg		997	1205	1605	1929	2130	1968	1852	1725	1607	1437	1123	886
	RadStd		76	149	125	157	135	144	122	124	105	120	89	85

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	+0.61	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-171	+286	+182

Nomenclature: See separate page